Calc 1 – Differentiation Rules

Worksheet A

1. Differentiate $f(x) = (3x^2 + 1)(x^3 - 2x)$ (product rule).

2. Differentiate $g(x) = \dfrac{\sin x}{1 + \cos x}$.

3. Differentiate $h(x) = \cos(x^3 + e^x)$.

4. Use implicit differentiation to find dy/dx for $x^2 + xy + y^2 = 7$.

5. Use logarithmic differentiation to find y' for $y = x^{\sin x}$ $(x > 0)$.

Calc 1 – Differentiation Rules

Worksheet B

1. Differentiate $f(x) = x^2 e^x \ln x$ (extended product rule).

2. Differentiate $g(x) = \dfrac{\sqrt{x^2+1}}{x}$.

3. Differentiate $h(x) = \tan\big(\sqrt{1+x^2}\big)$.

4. Use implicit differentiation to find the slope of the tangent to $\sin(xy) = y$ at $(0,0)$.

5. Use logarithmic differentiation to find y' for $y = (\ln x)^x$ (where defined).

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Answer Key (Worksheets A & B)

Worksheet A.

1. $f' = 6x(x^3 - 2x) + (3x^2 + 1)(3x^2 - 2) = 6x^4 - 12x^2 + 9x^4 - 6x^2 + 3x^2 - 2 = \boxed{15x^4 - 15x^2 - 2}$.

2. $g' = \frac{\cos x(1+\cos x) - \sin x(-\sin x)}{(1+\cos x)^2} = \frac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2} = \frac{1+\cos x}{(1+\cos x)^2} = \boxed{\frac{1}{1+\cos x}}$.

3. Chain rule: $h' = -\sin(x^3 + e^x) \cdot (3x^2 + e^x) = \boxed{-(3x^2 + e^x)\sin(x^3 + e^x)}$.

4. Differentiate: $2x + y + xy' + 2yy' = 0 \Rightarrow y'(x + 2y) = -(2x + y) \Rightarrow \boxed{y' = -\frac{2x+y}{x+2y}}$.

5. $\ln y = \sin x \ln x \Rightarrow y'/y = \cos x \ln x + \frac{\sin x}{x} \Rightarrow \boxed{y' = x^{\sin x}\Big(\cos x \ln x + \frac{\sin x}{x}\Big)}$.

Worksheet B.

1. $f' = 2x \cdot e^x \ln x + x^2 e^x \ln x + x^2 e^x \cdot \frac{1}{x} = \boxed{xe^x(2\ln x + x\ln x + 1)}$.

2. Rewrite $g = (x^2+1)^{1/2} \cdot x^{-1}$. Quotient or chain: $g' = \frac{x \cdot \frac{x}{\sqrt{x^2+1}} - \sqrt{x^2+1}}{x^2} = \frac{x^2 - (x^2+1)}{x^2\sqrt{x^2+1}} = \boxed{-\frac{1}{x^2\sqrt{x^2+1}}}$.

3. $h' = \sec^2\!\big(\sqrt{1+x^2}\big) \cdot \frac{x}{\sqrt{1+x^2}} = \boxed{\frac{x}{\sqrt{1+x^2}}\sec^2\!\big(\sqrt{1+x^2}\big)}$.

4. Differentiate: $\cos(xy)\,(y + xy') = y'$. At $(0,0)$: $1 \cdot (0 + 0) = y'$, so $\boxed{y'(0) = 0}$.

5. $\ln y = x\ln(\ln x) \Rightarrow y'/y = \ln(\ln x) + x \cdot \frac{1}{\ln x} \cdot \frac{1}{x} = \ln(\ln x) + \frac{1}{\ln x}$. Thus $\boxed{y' = (\ln x)^x\Big(\ln(\ln x) + \frac{1}{\ln x}\Big)}$.